MEDICAL DATA SECURITY USING BLOCKCHAIN

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***Abstract*—In general, hospital procedures require a large number of medical reports, which are an essential component of the process. Hospitals are now increasing their business by integrating pathology and other test laboratories into the hospital for efficient and speedy reporting, as well as increased income. Hospital operations cover a wide range of functions, from patient admission and management to hospital expenditure management. This, together with additional services such as pathology and pharmacy management, adds operational complexity and makes it harder to track. To address this issue, we employ blockchain technology to keep track of every single transaction with 100% authenticity using the hyperledger idea. All transactions are encrypted and saved as blocks to enable authentication over a network of computers rather than a centralized server. Furthermore, we employ the hyperledger idea to associate and preserve all of the medical papers linked with each transaction, including the date stamp. This enables for the authentication of each report, which will be identified if edited by anybody.**

***Index Terms*—Block chain in Healthcare, Medical Reports, Encryption, Authentication, Pathology, Pharmacy Management.**

I. INTRODUCTION

In the ever-changing world of healthcare, hospitals play a critical role in providing excellent medical services to patients. Hospital operations have become increasingly complex, involving everything from patient admission and management to the integration of pathology and test labs on hospital grounds. This connection not only allows for more effective and timely report preparation, but it also helps to increase and diversify the hospital's business portfolio. However, the development of hospital services, particularly the inclusion of pathology and pharmacy management, has added a layer of complexity that presents operational transparency and tracking issues. To overcome these difficulties, the use of blockchain technology emerges as a game-changing alternative, altering how hospitals manage and validate their operations. Block chain technology, which uses the unique notion of a hyperledger, provides a decentralized way to record-keeping, significantly changing the previous dependence on centralized servers. This decentralized computer network assures that all hospital transactions, from patient admissions to pathology and pharmaceutical administration, are properly logged and encrypted. The end result isa system that ensures 100% authenticity, creating trust in the hospital's operations. Furthermore, the use of the hyperledger idea goes beyond transactional data to include the organization and storage of critical medical papers associated with each transaction. Each document is securely time stamped, resulting in an immutable record that not only confirms the validity of medical reports but also provides a strong barrier against unauthorized changes. This degree of openness and security improves the integrity of the hospital's operations, encouraging a dependable and responsible healthcare system. In essence, the incorporation of blockchain technology, specifically the hyperledger concept, represents a transformative step forward for hospitals, ensuring not only the seamless management of diverse operations but also the authenticity and security of medical transactions and associated documents.[11]

II. LITERATURE REVIEW

On the grounds that wellbeing is a particularly significant part of an individual's life, medical services is one of India's biggest and quickest extending business areas. Starting around 2016, India's medical care business has developed at a build yearly development pace of practically 22%, and it is anticipated to arrive at USD 372 billion by 2022. It creates huge volumes of information in many structures, including individual wellbeing records. There is a critical requirement for further developed information the executives to make it interoperable while likewise tending to privacy issues. Patients, emergency clinics, safety net providers, and the public authority are the four essential partners in the patient clinical records cycle. Until date, electronic answers for this issue have been concentrated, using a client-server plan. Notwithstanding, such an incorporated plan has critical issues, including dangers of disappointment, attacks, and abuse of power. For this situation, a blockchain-based framework

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offers a better choice due than successful encryption, approval components, and interoperability. In our methodology, we propose a proof of idea answer for the main issue utilizing Ethereum blockchain innovation and a couple of extra devices.[1]

Blockchain innovation has exhibited surprising adaptability lately, as an extensive variety of industry areas tracked down strategies to integrate its capacities into their tasks. While most of the consideration has been on the monetary administrations industry, different activities in other help related spaces, as medical services, propose that this is evolving. This examination centers around the few starting focuses for Blockchain innovation in the medical care business. This paper utilizes models from public medical services organization, client situated clinical exploration, and drug falsifying in the drug area to exhibit the expected effects, objectives, and ramifications of this troublesome innovation.[2]

Light Chain is the primary DHT-based blockchain with logarithmic message and memory intricacy. In this demo paper, we exhibit the product engineering of our open-source Light Chain execution, as well as another organization procedure for the full Light Chain framework on a solitary PC, with the objective of reproducible results.[3]

Current medical services frameworks are portrayed as being incredibly convoluted and costly. Further developed wellbeing record the board, the utilization of insurance agency, and blockchain innovation can all assist with diminishing this. Blockchain was at first introduced to empower dispersed records of cash exchanges that didn't depend on concentrated specialists or monetary associations. Forward leaps in blockchain innovation have brought about upgraded exchanges including clinical records, insurance charging, and savvy policies, considering extremely durable information access and security while likewise offering a dispersed data set of exchanges. One critical advantage of utilizing blockchain innovation in the medical care industry is that it can work on the interoperability of medical services data sets, giving patients more admittance to their clinical records, gadget following, solution data sets, and emergency clinic resources, as well as the whole life pattern of a gadget inside the blockchain network. Admittance to people's clinical history is basic for precisely endorsing prescription, and blockchain can possibly fundamentally further develop the medical care administrations system. This paper investigates a few answers for working on current constraints in medical care frameworks utilizing blockchain innovation, including structures and devices for estimating framework execution, like Hyperledger Texture, Writer, Docker Holder, Hyperledger Caliper, and the Wireshark catch motor. Moreover, this work presents an Entrance Control Strategy Calculation for expanding information openness between medical care suppliers, as well as supporting the demonstrating of settings for executing the Hyperledger-based electronic health record (EHR) sharing framework, which utilizes the chain code thought. Execution boundaries in blockchain networks, like idleness, throughput, and Round Trip Time (RTT), have additionally been improved to come by improved results. Not at all like standard EHR frameworks, which utilize a client-server engineering, the recommended approach utilizes blockchain to further develop productivity and security. Networks should manage always showing signs of change and obscure worries. This can be tended to by anomaly based neural network (NN) intrusion detection systems (IDSs) that have been appropriately prepared and tried. Thusly, the IDS ought to be tried to perceive how effectively it identifies these assaults. Evaluating NN IDSs might be troublesome and complex. One ought to have the option to decide how rapidly the IDS consolidates and how really it performs (how much of the time it flops and how every now and again it recognizes issues). This paper examines the different strategies utilized by experts to prepare and assess their IDS models. It was likewise found that the data utilized can impact how effectively the NN IDS models are delivered and tried.[4]

The quick ascent of the Internet of Vehicles (IoV) has presented huge issues for immense information stockpiling, smart administration, and data security all through the entire frame- work. The standard concentrated organization answer for IoV has issues adapting to ongoing reactions. The blockchain, as a powerful innovation for decentralized disseminated capacity and security the board, has proactively shown huge advantages in its application to Bitcoin. In this review, we take a gander at how blockchain innovation might be utilized to vehicle organizing, to be specific the conveyed and safe capacity of gigantic information. We lay out many sorts of hubs for vehicle organizations, like vehicle and side of the road, and make a few sub-blockchain networks. In this review, we give a model to the outer transmission of vehicle blockchain information, trailed by definite hypothetical examination and mathematical discoveries. This article has shown how blockchain might be utilized to drive future vehicle organizing applications.[5]

Electronic health records (EHRs) are to a greater extent used than customary paper-based clinical records because of their proficiency, security, and capacity to diminish information overt repetitiveness. In any case, EHRs actually have unfortunate interoperability, and protection concerns stay ignored. Blockchain, a disseminated record innovation comprised of encoded blocks of information organized in chains, can possibly address EHRs' deficiencies with regards to interoperability and security. In this article, we make sense of EHRs and blockchain innovation, as well as give various conventional blockchain-based strategies for further developing EHR interoperability and security. We then examine recent concerns in information the executives productivity, access reasonableness, and framework trust. Here, we propose continuous examination needs in wellbeing informatics, information sciences, and morals to develop EHRs utilizing blockchain innovation. Blockchain-based EHR frameworks should deal with the critical aberrations in medical services assets, the gigantic carbon impression of processing requests, and the expected doubt of wellbeing professionals and patients that might result from the boundless utilization of blockchain innovation.[6]

Electronic trade of clinical imaging information is an essential part of current medical services frameworks, but

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current foundation for cross site picture move depends on confidence in outsider middle people. In this paper, we research the blockchain thought, which permits members to agree without contingent upon a focal power. We make a framework for cross space picture sharing that utilization blockchain as a dispersed information stockpiling to make a record of radiological examinations with patient-characterized admittance privileges. The blockchain structure is exhibited to forestall outsider admittance to safeguarded wellbeing data, meet a few measures for an interoperable wellbeing framework, and effectively adjust to regions other than clinical imaging. The structure's overall weaknesses incorporate the intricacy of the protection and security ideas, as well as a questionable administrative climate. At long last, the enormous scope common sense of such a technique should be demonstrated, and this not entirely set in stone by various standards that we will go over top to bottom.[7]

III. Methodology

In this section, we describe the basic overview of the System Architecture in Fig.1.

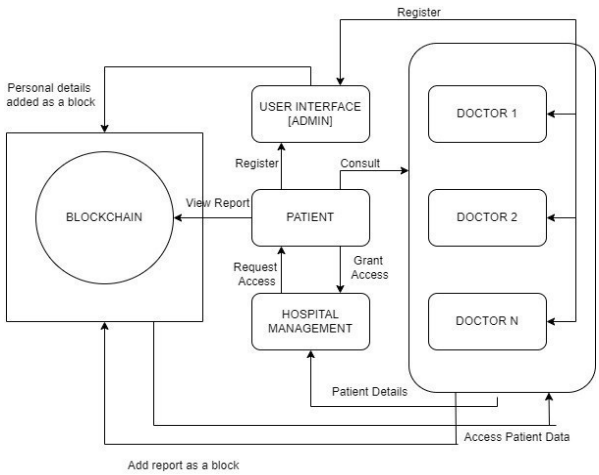

Fig. 1. The System architecture provides an overview of these major components and their roles within the system.

As we can see Fig. 1 represents the major key components used in this project which includes :

- Registration: Where new patients enter their details.
- User Interface: How patients interact with the system.
- Doctors: Healthcare providers within the system.
- Blockchain: The secure, distributed technology storing patient data.
- Hospital Management: The administrative side of the hospital.
- Patient Details: Medical records stored on the blockchain.
- Grant/Request Access: Controls who can view patient records.
- Add Reports: How new medical reports are attached to patient records on the blockchain.

In earlier works[9], the present medical data system has issues with privacy and accessibility. To improve security and enable sharing, we present a method that combines Hyperledger Fabric with attribute-based access control. This uses blockchain and smart contracts to enable tamper-proof storage of medical information while also providing dynamic access. The addition of IPFS technology resolves blockchain storage difficulties. Experimental findings show that our technique is effective in providing safe, integrity-preserving, and high- throughput access to medical information in the age of big data health care informatization.

A. Drawbacks:

1) Integrating Hyperledger Fabric and ABAC may need complex installations.
2) Users and administrators may struggle to adjust to new technology.
3) Running blockchain and IPFS can need significant computing resources.
4) Compatibility concerns may occur when merging with existing medical systems.

Introducing a breakthrough approach for hospitals, our suggested blockchain-based solution, which leverages the hyperledger idea, provides operational transparency and security. This decentralized technique ensures complete accuracy in tracking patient admissions, pathology, and pharmaceutical transactions. Hyperledger goes beyond transactions, securely timestamping and connecting critical medical papers, resulting in an immutable record. This radical leap increases general trustworthiness, resulting in a dependable and responsible healthcare environment.[10]

B. Benefits:

1) A transparent block chain promotes confidence in healthcare operations and transactions.
2) Hyperledgers provide safe, unalterable records for patient data and transactions.
3) The decentralized model promotes responsibility throughout the healthcare sector.
4) Hyperledger securely timestamps crucial medical papers to ensure their legitimacy.
5) The revolutionary method promotes dependability, creating a trustworthy healthcare atmosphere.

C. Modules:

As mentioned in Fig.2 Module Architecture,

1) Admin Login:
    - Add Doctor/Hospital Details: Admin can add information about physicians and hospitals to the system.
    - Admin can view doctor and hospital details saved in the system.
2) New Patient Signup:
    - Allows new patients to establish accounts in the system.
3) Patient Login:
    - Patients can share their health and medical records with approved parties.
    - View Health Report: Patients can view their own health reports.
    - View Hospitals: Offers patient information about hospitals.

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4) Doctor Login:
   - Read Patient Report: Doctors can access and read their patients' health records.
   - Prescription Note: Allows doctors to prescribe drugs and give medical advise.[16]

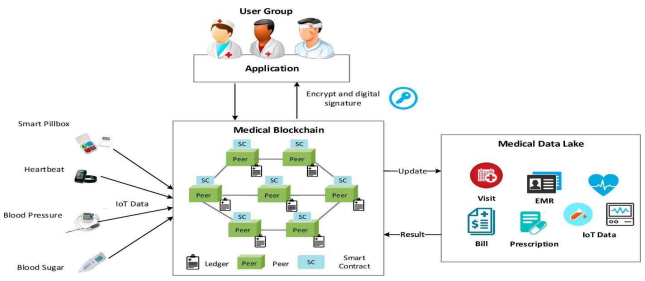

Fig. 2 outlines the module architecture, covering the core functionality available to admins, patients, and doctors within the healthcare management system's implementation

IV. IMPLEMENTATION

The suggested healthcare management system has separate features for administrators, patients, and doctors.

1) Admin Module:
   The administrator may access two critical features in the Admin Login section: 'Add Doctor/Hospital Details' and 'View Doctor/Hospital Details.' The administrator uses the former to enter critical data on doctors and hospitals into the system, allowing for more effective management. The latter allows the administrator to monitor and manage the stored information about doctors and hospitals.

2) Patient Module:
   New Patient Signup enables users to register in the system and begin their healthcare journey. Once enrolled, patients may access 'Share Health/Medical Report,' which allows them to securely send their health reports to approved institutions. Additionally, 'View Health Report' gives patients easy access to their own health information, fostering openness. View Hospitals provides patients with useful information about various hospitals.

3) Doctor Module:
   The Doctor Login area includes 'View Patient Report,' which allows doctors to examine the health records of their assigned patients. The 'Giving Prescription Note' capability allows doctors to prescribe drugs and provide medical advise, hence expediting the patient care process.

This complete system encourages collaboration among healthcare stakeholders, resulting in more efficient information flow and improved patient care. The deployment includes strong security measures to protect sensitive health data, as well as adherence to regulatory standards for healthcare information management. Furthermore, user-friendly interfaces for patients, physicians, and administrators help to provide a smooth and efficient healthcare management experience.[12]

V. EXPERIMENTAL RESULTS

Block chain Interaction:

- The application interacts with the blockchain through HTTP providers, connecting to a specified Ethereum node.
- Smart contracts deployed on the blockchain define the structure and behavior of healthcare-related data, ensuring that it is stored securely and can be accessed transparently by authorized parties.[14]

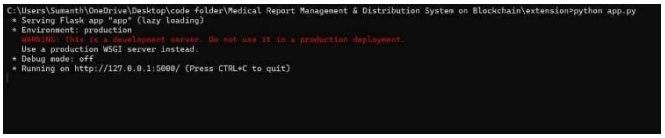

Fig.3.The Flask application interacts with the blockchain by using the Save_Data_BlockChain function, which leverages the Web3 library to connect to the Ethereum blockchain.

Link to application:
The flask application that interacts with the blockchain using the Save_Data_BlockChain function, which utilizes the Web3 library to connect to the Ethereum blockchain as shown in fig.3.

Adding doctors and hospitals:
The process of adding doctors and hospitals is managed via the /AddDoctorAction route by admin. Here, the data is stored on the blockchain using the saveDataBlockchain function as shown in fig. 4.

Fig. 4. The admin page functionality is to register new doctors and hospitals into the system which is accessible to administrative users

View Hospital/Doctor Details:
Hospital details can be viewed through the /View Hospital Details route, which retrieves data from the blockchain using the read Details function as shown in fig. 5.

Fig. 5. The view Hospital Details route that displays the directory of hospitals and doctors affiliated with the healthcare network

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Upload Details of Patient :
Patient health records are submitted through the Add Health Action route and saved to the blockchain using the saveDataBlockChain function as shown in fig. 6.

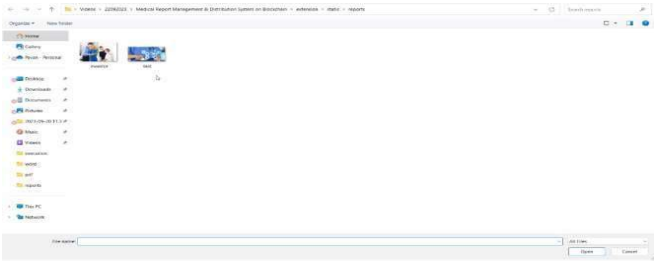

Fig .6. The medical records and health information of patients are entered into the system via the 'Add Health Action' route.

The blockchain ensures that health records of patient are tamper proof and can be accessed securely by authorized parties such as doctors or administrators.
View Patient Details:
Patients and authorized individuals can access health records via the /ViewPatientReport and /ViewHealth routes. These routes retrieve data from the blockchain using the readDetails function as seen in fig. 7.

Fig. 7. Patients and those with proper authorization can retrieve and view health records and medical reports by accessing the '/ViewPatientReport' and '/ViewHealth' routes

Adding Prescription Details:

- When a patient requests a prescription via the /Prescription route, the data is sent to the Flask application.
- The Flask application then communicates with the blockchain using the saveDataBlockchain function, leveraging the Web3 library to establish a connection with the Ethereum blockchain.

- Prescription data is securely recorded on the blockchain using smart contracts, which specify functions for both setting and retrieving prescription details as shown in fig. 8.

Fig .8. The patient sends a prescription request through the '/Prescription' route, it is directed to the responsible doctor for review and approval.

Fig. 9. The doctor issues a prescription for a patient by using the '/PrescriptionAction' route, the prescription information is successfully added to patient's medical records.

- Similarly, when a doctor creates a prescription via the /PrescriptionAction route, the prescription data will be saved to the blockchain as shown in fig. 9. which guaranteeing immutability and transparency.

VI. CONCLUSION

To summarize, the integration of blockchain technology, particularly the Hyperledger concept, has proven to be a transformative solution for addressing the complex issues encountered in modern hospital operations[13]. The integration of various operations within a healthcare ecosystem, including patient admission and management, pathology, and pharmacy services, has significantly increased operational complexity. However, blockchain technology has emerged as a reliable foundation for streamlining these activities while enhancing efficiency and security. Hospitals utilize blockchain to ensure the trustworthiness and legitimacy of all transactions, including clinical reports. The decentralized nature of blockchain, distributed across a network of computers, provides unprecedented protection, reducing the risk of unauthorized alterations. The use of Hyperledger strengthens this security paradigm by ensuring that all connected medical documents are properly maintained and timestamped. This method allows for seamless transaction tracking and also creates an unalterable record of each patient's complete medical history. Furthermore, the use of blockchain technology improves hospital expenditure management and promotes a more flexible and responsive healthcare system. As hospitals evolve into comprehensive healthcare hubs, blockchain serves as a cornerstone, offering a transparent, secure, and reliable foundation for managing the intricate network of activities. In summary, the utilization of blockchain technology in hospital operations represents a significant step toward a future in which authenticity, transparency, and efficiency converge to reshape healthcare management norms.[15]

VII. FUTURE WORKS

The future of blockchain technology in healthcare goes beyond operational efficiency to include sophisticated applications like customized treatment, real-time data analytics, and interoperability across healthcare systems. Integration with new advancements, for example, AI and the IoT has incredible

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commitment for working on persistent consideration, treatment results, and exploration collaboration, and also we can work on development of pharmacy and laboratory modules to seamlessly integrate into our medical report management & distribution system[17]. By integrating these modules, we aim to create a cohesive platform that optimizes the healthcare experience for all users involved. As the healthcare sector evolves, blockchain's role is expected to grow, providing creative solutions for data exchange, patient consent management and supply chain efficiency. This trajectory establishes blockchain as a foundation for designing the future generation of healthcare ecosystems, encouraging cooperation and assuring patient-centric, digitally enabled healthcare delivery.[18]

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